THE INSTRUMENTS

BY

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RESEARCH AND THEORY
WHAT A MEASUREMENT CAN SEE, WHAT IT CANNOT, AND HOW EACH NULL WAS DRAWN

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ABSTRACT

A battery of null results is worth only the sensitivity of the battery. The digits of pi pass every test assembled here against SHA-256, uniform bit shares, flat autocorrelation, full algebraic degree, no co-variation and incompressibility, and the digits of pi have a generating program of about a kilobyte. What the battery establishes about SHA-256 is therefore a set of bounds with domains: bits unbiased to about four parts in a million over 550 billion evaluations, no linear approximation surviving out of sample, and full degree by round eight. Beside those bounds sit a real signal in the block header, where the version-rolling bits give away 1.82 bits of collision entropy, and a support measurement that needs no statistics: at round seven, 214 of 256 output positions are forbidden to a single input bit.

Fifteen failure modes of the apparatus are listed with the subtraction that removes each. Every instance of every mode makes a result look better than the truth. Five have a mechanical detector, five a partial one and five none. The rules that follow are to read the interior where it is readable, run a positive control, draw a bar by resampling and never derive it, match the null to the nuisance parameters, normalize until the baseline is flat, measure at two widths, and suspect the instrument before the object.

Seven transforms carry the work, and each pays only where the object has fewer coefficients than points. A hash is built to have as many, and no transform here shortens one.

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1

WHAT THE INSTRUMENTS CANNOT SEE

This chapter states what is measured about SHA-256 and the block header, at what scope, and which instruments cannot see what they are pointed at, because a battery of null results is worth only the sensitivity of the battery.

One finding outweighs every null result about SHA-256 here: **the digits of pi pass every test in the battery, and the digits of pi have a generating program of about a kilobyte.**

That is not a remark about pi. The battery checks for uniform bit shares, flat autocorrelation, full algebraic degree, no co-variation between output positions, and incompressibility to LZMA. A sequence with a short generating program sails through all of it. Reporting *no structure found* on the strength of it says less than it appears to.

1.1 FIVE INSTRUMENT FAULTS

All five run in the same direction, each producing an apparent finding, and each is detected by a result being too clean and not by care beforehand.

A statistic read off the top of a sort. The deepest timestamp reversals appear to cluster near 280 seconds. They are the eight largest values of a sorted list, which are close together by construction. The full population shows no band.

Three noise floors derived instead of drawn. A shuffle of hour *labels* returns identical counts every draw and prints a standard deviation of 0.00. A per-pool phase bar set at two Poisson floors, for a statistic that is a maximum minus a minimum over twenty-four bins and spans three and a half to four and a half by construction, sits below chance: every pool clears it, and twenty random trough hours read as geography. A window-scatter floor from chi-square's variance carried through a logarithm gives 0.0582 where shuffling gives 0.1015.

A generator with a period. Survey arms built as $at \times (2\,arm + 37) + 11 + 97\,arm \bmod 256$ are linear in the arm index, and both terms wrap together every 256 arms: arm 0 and arm 256 produce byte-identical headers. Nine hundred thirty one files hold two hundred fifty six distinct arms, up to four copies each, and the duplication inflates the summed squared z from 256 to 942. That reads as a thirty sigma discovery and reproduces the duplication factor to within one percent. Hashing each header from its arm index, instead of stepping it, removes the period.

A test whose answer is geometry. Reading a state through a mirrored helix appears to test the handedness of SHA-256, which is chiral by design: six rightward rotations and no leftward one. But reflecting a point set negates every longitude, conjugating every coefficient and negating

every phase, for any data whatever. The result is fixed before the data is consulted and comes back at 10^{-16}, which is float64's floor.

A scalar standing in for a structure. Per-degree power is invariant under all of SO(3), and reporting it alone implies that magnitude and phase separate cleanly. They separate only on rotations about the reading axis. Under a general rotation the coefficients mix, magnitude redistributes across orders inside a degree, and the summed power is preserved because that mixing cancels and not because nothing moved. Summing over orders makes the mixing invisible.

1.2 WHAT IS ESTABLISHED, AT ITS REAL SCOPE

On SHA-256, bounded and not proven flat. Across 550 billion evaluations spanning 256 genuinely distinct headers, the output bits are individually unbiased to about four parts in a million and carry no co-variation a permutation null cannot produce. A Walsh-Hadamard sweep over all 262,144 input masks per output bit, the linear approximations of linear cryptanalysis [15], finds no linear approximation surviving on nonces it did not see; every in-sample maximum sits below the null's own expected maximum. The algebraic degree in the nonce bits saturates by round eight, and no transform has fewer coefficients than points to ride. Reversing every rotation gives a function statistically identical to the standard one [9]: avalanche [20] means differ by a third of a standard error, and both reach half the output at round eighteen. The avalanche criterion over every combination of the seven sub-functions is measured in [19], which finds no attack arising from combining them. Handedness is cosmetic, and the rotation amounts carry the diffusion.

Every one of those is a bound with a domain, and not an absolute. A finite observation of an unbounded input space can bound an effect and can never establish absence.

On the block header, a real and large signal. The sixteen bits BIP320 [5] reserves for version rolling, which the Stratum extension of BIP310 [4] negotiates, give away 1.82 bits of collision entropy, 588 times the floor chance produces at that width and depth, and 45.7 percent of what the field could carry. Per-bit shares run from −5 to −51 standard errors against a swept counter's prediction, and the whole distribution is bimodal with spikes on exact powers of two: conventions, and not a sweep. The leak also moves, at $p = 0.010$ against a shuffle null, which reads as software being deployed and retired inside a forty eight day window.

On the clocks, a physical effect cleanly separated. Timestamps in 2.98 percent of blocks precede their own parent's, by up to 395 seconds. The clocks tick correctly: low-bit shares reach 3.10 against a calibrated null of 3.35, and residues mod ten give chi-square 9.9 on nine degrees of freedom. The mining process is a clean Poisson at sd/mean = 0.9879. The effect is therefore offset, and neither drift nor granularity, and the magnitudes are four orders past anything NTP produces.

And the separation that ties them together. The tilt lives in the fields miners *choose* and is absent from the field the function *produces*. Across 178 testable digest positions, conditioned per block on its own target, the loudest reads +2.78 standard errors against a 3.22 bar. Same eighty bytes, same blocks, same miners.

1.3 WHAT HAS NO INSTRUMENT

A one-way function is the gap between a description of about a thousand bits and a search costing 2^{48}, and nobody has proven one exists. SHA-256's hardness is conjectural. Every measurement above assumes the gap and confirms its local consequences, which is circular, and the pi control exposes the circle.

Kolmogorov complexity [13] is uncomputable, and no tool decides it. A compressor gives an upper bound: compression proves structure, failure to compress proves nothing, because a compressor hunts repetition and a generating program is not repetition. Measured directly, pi compresses to 1.0010 against random's 1.0001. The positive control fails completely, and any claim of the form *it does not compress, therefore no structure* is a statement about zlib.

What separates pi from noise is neither distribution nor compression. It is reproducibility: an independent method computes the same digits, and agreement between two programs sharing no arithmetic demonstrates a generating program. That is a predictive test, and the predictive test for a hash is the linear approximation search, the one axis above that pi would fail.

1.4 READING THE INTERIOR INSTEAD OF THE BOUNDARY

Every measurement above reads a digest, the boundary of the computation. That is the right shape for a physical system, where the interior cannot be reached and has to be inferred from what escapes. It is the wrong shape here, because nothing about this interior is hidden. The schedule words, the eight state words at every round, and the six values built inside a round are all readable at every instant. Those six are s_1, ch, t_1 on one chain and s_0, maj, t_2 on the other. Nothing forces a boundary reading here.

Read that way, one flipped input bit shows something the digest cannot.

intermediate	first round it moves at all
s_1, t_1, s_0, t_2	2
ch, maj	3

The nonlinear functions lag the linear ones by a round, and the lag persists: at round seventeen the linear values sit at 15.5 to 16.0 bits of difference while Choose and Majority sit at 14.3 and 14.8.

The mechanism is in their definitions. Choose is $(e \wedge f) \oplus (\neg e \wedge g)$. Where f and g *agree* at a bit position the result does not depend on e there at all. Majority cannot be moved by one of three inputs either. The nonlinear functions therefore absorb the first arrival and pass it on only once a second input has been reached. A digest sums all six paths into one number and cannot say which carried what.

1.5 THE RULES, WHICH ARE A DIFFERENT KIND OF STATEMENT

Everything above is a frequency, and a frequency needs a null, a bar and an argument. The support does not. For a single-bit input difference, an output position either moves or it never does,

and a position that never moves in six hundred excitations is not rare. It is forbidden, and every trajectory requiring it is ruled out outright.

rounds	reachable of 256	forbidden	share
1	0.0	256.0	100.0%
3	7.3	248.7	97.1%
5	24.4	231.6	90.5%
7	42.0	214.0	83.6%

At round seven, 214 of 256 positions are forbidden to a single input bit. The support grows about nine positions a round after the third, putting the grammar's closing near round thirty, and that is consistent with the avalanche independently reaching 126 bits moved by round twenty. The function therefore spends roughly its first thirty rounds before every input can reach every output, and its remaining thirty-four past the point where the support constrains anything.

This is the sharpest result here and the only one needing no statistics to stand up. Its limit is equally sharp: forbidden transitions prune trajectory *searches*, as differential cryptanalysis does. Mining evaluates forward and searches no trajectories. The grammar does not help there.

1.6 TWO RESPONSE MEASUREMENTS AND A LEAD, READ CORRECTLY

The faults here share a shape with the five above.

Susceptibility does not clear. The loudest cell of the real 512×256 table reads -4.33 standard errors against a null table's loudest of -4.11. A maximum over 131,072 cells is itself a random variable with real spread, and one draw of it is not a bar. Calling that a clearance is a derived threshold set too low.

The pair response has no power at all. Flipping two input bits and comparing against the exclusive-or of the two single differences gives 128.012 bits, and the null arm gives 127.938. Both are pinned at 128, which is half of 256 and therefore complete decorrelation. The statistic is saturated in both arms and cannot discriminate. Calling the excess *the nonlinearity* implies it carries information, and it does not.

A lead that dissolves at depth. At 1,200 samples the per-bit batch-to-batch swing has a loudest of 1.394 with the top swingers uniformly cold, which looks like a second channel invisible to a heat map. At 60,000 samples the null arm's own loudest is 1.3965, no bit clears it, and the correlation between swing and heat is -0.0671 inside a band of 0.1250. The anticorrelation is the artifact, and no signal remains.

1.7 THE TOOLS

tool	what it answers
compressibility with a pi arm	whether a null escapes a known short program
Walsh-Hadamard bias	every input mask by transform, and not a sampled few
handedness	SHA-256 against its mirror, rotations reversed
spectral invariance	invariance shown alongside the mixing it cancels
pooled arms	pooled depth and sign agreement, separated
agglomeration	co-variation, which every other test here is blind to
placement validator	whether a placement on the sphere is sound
corpus comparison	the clean survey against the contaminated chain
collision entropy	every reading in one comparable unit
watchdog	calibrated models instead of guessed thresholds, as in CFAR detection [18]
propagation	one flip through every interior value, not just the digest
support grammar	what is forbidden, needing no null
response	susceptibility, pair response, decay
anisotropy detector	flat baseline, drawn bar, names the angular scale
two-stage scanner	two stages over disjoint classes, validated before believed
scanner comparison	any new scanner proves itself on planted structure first
integer relation search	integer relations by lattice reduction [8], gated both directions
formula search	BBP-type formulas, gated by rediscovering pi's

All eighteen are in the orior repository, `https://github.com/dstroy0/orior`.

1.8 THE ONE MEASUREMENT THAT CAN RETURN A DISCOVERY

Everything above returns a bound or a null. The relation search returns a statement: a closed form either holds or it does not, and anyone can check it in an afternoon without repeating the work.

Its control is the 1996 BBP formula for pi [3]. Pointed at base sixteen, power one, stride eight, the pipeline returns an exact relation, and the known formula holds on the same values to 895 bits of 900, the remainder being the series truncation. The relation is not unique: Bailey's compendium [2] lists the zero relations, BBP-type sums that evaluate to zero, and notes that an integer relation search returns one unless they are excluded. The exact relations form a lattice, and the known formula is one member. Demanding a particular member is the wrong control and reports a correct answer as a failure.

One region is closed and should not be searched. Borwein, Galway and Borwein [6] prove that pi has no Machin-type BBP arctangent formula in any base $b \neq 2$ that is not a proper power, and they know of no constant lacking a Machin-type formula that has a BBP formula of any other form.

2

FAILURE MODES, AND THE SUBTRACTION THAT REMOVES EACH ONE

This chapter holds the apparatus to the same subtraction the data is subjected to. It lists fifteen failure modes, each with an instance, the detector that removes it and the detector's status.

This is not a checklist. It is a sieve, and the operation is subtractive: each detector removes a class of explanation for a number, and what is left after every subtraction is the residual. The residual is the finding. That is the same operation as the sigma-delta null [16], measurement minus what a null would have given, applied to the apparatus and not to the data.

Every instance of all fifteen modes makes a result look better than the truth. None errs toward a weaker finding. That asymmetry is itself the strongest reason to run the sieve instead of trusting a number.

Status is one of **built**, where a mechanical detector exists and runs over every measurement; **partial**, where a detector exists but is applied only where a defect is already suspected; and **open**, where the mode is known and nothing yet catches it.

2.1 ABSENCE READ AS MEASUREMENT (BUILT)

Measuring a quantity that structurally cannot depend on the input, and reporting the resulting zero or perfect agreement as a result.

Instances. Rotating the nonce, which enters at `W[3]` in round three, instead of the chaining value, reports a decay that never happens. Measuring through `W[17]`, which does not depend on `W[3]`, reports 100.000% agreement, which is zero equaling zero. Labeling rounds 1 to 3 *premise holds: no* is a verdict on a nonce that has not entered yet.

The subtraction. Exact reachability, computed and not sampled. Every state bit carries the set of input bits that can reach it, propagated through the real operations: rotation permutes positions, shift drops them, bitwise operations union bit by bit, and addition unions upward through the carry. Addition is the only widening operation, making this an over-approximation. A path may carry no influence, but **the absence of a path is exact**, the correct side to err on for a precondition.

What it settles. The nonce reaches no state bit until four rounds have run, and saturates both written words the moment it arrives:

rounds run	1	2	3	4	5
bits of `state[0]` reachable from `W[3]`	0	0	0	32	32

In the schedule, `W[16]` and `W[17]` carry **zero** bits of `W[3]` while `W[18]` carries all 32, correctly, since `W[18]` is the first word whose `W[t-15]` term is `W[3]`. Both of the first two instances are visible directly in those two tables, before the measurement instead of after it.

The control, run in the opposite direction. An analysis claiming reachability everywhere would pass an absence test trivially. Both halves are needed. Over 24,000 random flips at rounds one to six:

pairs the analysis called unreachable	21,257
of those, a flip that changed the bit	**0**
pairs the analysis called reachable	2,743
of those, a flip that landed	35.07%

One violation in the first row falsifies the analysis outright. The second row has to sit away from both extremes, and does.

A structural fact that comes free. Every one of the 256 state bits reads all 768 input bits from **round 21** on, the widest cone growing 256, 320, 352, 384, 416, thirty-two bits per round after the first two, which is one schedule word arriving each round. That number is exact.

2.2 HAND-DERIVED NULL (PARTIAL)

Comparing a measurement against an analytical reference derived by hand, where the derivation is wrong.

Instances. An expected worst of 3840 cells gives a 5.47 sigma keyhole that falls to 4.19 and then to nothing at 64× the samples. Overshoot modeled as geometric in nibbles produces a 7.27× excess; the target is not on a nibble boundary and the true figure is 1.0076. A max-statistic compared against a per-bin uniform produces 775,000× where the permutation null gives 184.5×, inflated by about 4,200.

The subtraction. $\Delta_{\Sigma}[M] = M(x) - \mathbb{E}_{\pi \sim \Pi_{\Sigma}}[M(\pi(x))]$. A permutation null cannot be derived wrongly because it preserves the carrier's own counts by construction and not by arithmetic.

Status. Built and applied to one statistic. The keyhole scan's expected worst and the corpus-size ratios are both hand-derived and neither is checked.

2.3 ASYMMETRIC TREATMENT OF THE TWO ARMS (OPEN)

Applying one operation to one member of a pair and a different operation to the other, leaving the difference in treatment inside the measured difference.

Instance. Adding K to one member and $\mathrm{rotr}(K)$ to the other produces a 25.8% survival that appears to overturn a measured zero. The same constant to both gives 0.000000%.

The subtraction. A swap test. Exchange the two arms and re-run. A statistic meant to be symmetric must return the same number; one deliberately asymmetric must say so and say why.

2.4 READING MORE OUTPUT THAN THE OPERATION WRITES (BUILT)

Instance. Measuring all eight output words when a SHA-256 round writes only `state[0]` and `state[4]`. The six untouched words agree trivially and the result reads $P = 1.00000000$.

The subtraction. A write mask, computed beside the read set, since it is the same question about what the circuit does. Every output word is assumed a copy of every input word, and each of 2000 random trials removes the pairings that failed; what is left after every trial is carried instead of written.

rounds run	1	2	3	4
words written	2	4	6	8
words carried	6	4	2	0
share of an eight-word read that agrees for free	**75%**	50%	25%	0%

An eight-word read after one round agrees on three quarters of its comparison before the round function is consulted. Reading only the written words needs no correction factor and no argument. It is the right read.

2.5 SAMPLE BELOW THE DETECTABLE EFFECT (OPEN)

Running an experiment whose expected count under the hypothesis is far below one, and reporting the absence of an effect as evidence.

Instance. $k = 8, 12, 16$ at 400 states, where $k = 16$ predicts 0.006 hits. $k = 4, 6, 8$ at 5000, with the detectable effect size printed alongside, can see the effect.

The subtraction. Every measurement prints its expected effect size and its detection floor *before* its result. A null then reads as *below the floor* and not as *absent*.

2.6 ESTIMATOR CEILING SET BY SAMPLE SIZE (PARTIAL)

Reporting a plug-in estimate whose value is fixed by how many samples were drawn and not by the distribution.

Instance. The Rényi [17] order one half plug-in is $2 \log_2 \sum \sqrt{p}$. Where every observed value is a singleton that sum is exactly $\sqrt{n}$, and the estimate reads $\log_2 n$ whatever the true entropy is. At 2,000,000 pairs the ceiling is 20.93 bits and the estimate reads 18.25.

The subtraction. Print the ceiling next to the estimate. Where they are close the estimate is mostly sample size and is not a measurement.

Status. Done for the Rényi estimates. Every other sparse-sample estimator is unchecked, and Good-Turing coverage [12] is the obvious place to look.

2.7 DEGENERATE PAIRS (PARTIAL)

Constructing a pair meant to differ and failing to check that it does.

Instance. Exclusive-or at positions that are not distinct flips bits back and leaves some pairs identical. The result prints *smallest output distance: 0 bits*, which reads as a collision, directly above prose saying nothing came close.

The subtraction. Count and report the pairs whose intended difference is zero. A nonzero count reports a construction defect and carries no result.

2.8 OPTIMIZER-DEPENDENT RESULT (BUILT)

A compiler folds, contracts, reassociates and hoists. A folded measurement is indistinguishable from a computed one by reading the output.

Instance. A whole hashing implementation that is a constant function under optimization, and three performance ratios that move with the build.

The subtraction. Build every measurement three ways, `-O0`, `-O2`, and `-O2 -ffp-contract=off`, and compare the outputs with stopwatch lines dropped.

Result. 19 of 22 judged measurements are byte-identical across all three arms. The three that move are all ratios of measured throughputs: extranonce cost 11.1× / 28.3× / 23.4×, expansion speedup 0.932× / 0.395× / 0.256×, delivery search-over-verify 1.08×10^{19} / 5.46×10^{19} / 6.15×10^{19}.

What it does not close. That a folded constant is folded *correctly*. For the Rényi measurement that is settled separately by a reimplementation of every printed prediction in another language against another libm. Nothing else has that check.

An alignment fault under the unoptimized build. The known-answer test and one measurement produce no `-O0` output because they segfault there. The trace:

```
vmovdqa %ymm0,-0x20(%rbp)   rbp = 0x5fe0b0, target 0x5fe090, which is 16 mod 32
```

`vmovdqa` is the *aligned* 256-bit move and requires 32. MinGW-w64 GCC at `-O0` hand-aligns a pointer for one over-aligned `__m256i` local, the emitted `add $0x1f; shr $0x5; shl $0x5` being visible in the prologue, and then places another at a fixed offset from a frame pointer it never realigned. `-mstackrealign` changes the prologue not at all. **The fault is in the compiler, a known defect of 32-byte stack alignment on 64-bit Windows [10], and no source change avoids it**; the instruction is available and `-O2` runs it happily.

The resolution is that the `-O0` arm drops `-mavx2`. The hashing core gates its vector arm on `__AVX2__` and the `#else` arm defers to the scalar reference. The unoptimized build is therefore the reference arm, the arm a fold audit wants, since it is the arm whose arithmetic the statistics read. The known-answer test then runs `-O0` clean: **24 run, 0 failed.**

And what that costs. For the two measurements exercising the vector arm, the `-O0` column tests different code from the other two columns. Agreement there is evidence about the reference arm and not about the vector arm. The vector arm is covered instead by the known-answer test asserting it equal to the reference on the FIPS 180-4 vectors [9], the genesis block, block 125552 and 1000 consecutive real headers. That is a strong check, and it is a different check, and reading the audit table as though all three columns meant the same thing for those two rows would be the kind of substitution this sieve exists to catch.

2.9 PROSE CONTRADICTING ITS OWN TABLE (OPEN)

Instance. Prose saying the two *agree often* directly above a column reading 0.0000%.

The subtraction. Every prose claim names the number it rests on. The two can then be compared without re-deriving the claim.

2.10 CONFLATED QUANTITIES (OPEN)

Instance. A characteristic, a product over one chosen intermediate path, treated as a differential, the enumeration over all intermediates. Only the two zeros in such a table are informative.

The subtraction. Naming discipline. Two quantities that differ get two names, and the name records which one is meant.

2.11 PARTIAL STANDING IN FOR THE WHOLE (PARTIAL)

Instances. Reduced rounds in five measurements, one output word of eight, a 64-bit prefix of a 256-bit digest, one frame, one constant, chunk two without the header, sampling in place of enumeration. Every one of these is a projection reported as the thing.

The subtraction. Lead with the whole, and label a partial as partial in the same breath. The Rényi measurement enumerates all 2^{32} nonces instead of sampling, and reads the whole 256-bit digest.

2.12 MAX-STATISTIC AGAINST A PER-BIN NULL (PARTIAL)

Reporting the largest of many bins against the expectation for one bin.

Instance. The 775,000× above, which the permutation null puts at 184.5×.

The subtraction. The permutation null over the maximum itself [21], or an extreme-value reference. Drawing n samples into r bins gives an expected maximum well above n/r, and that expectation is the reference.

2.13 ARGMAX OVER A TIED SET (PARTIAL)

Selecting *the most common* value where many values share the top count. The sort's tiebreak selects, and the data does not.

Instance. A walk through delta space steered by the most common outgoing delta. Counting how many share that maximum:

round	0	1	2	3	4	5	6
deltas tied at the peak	1	**12**	**21**	**19**	**17**	**29**	**17**

From round one onward there is no single most common delta. Between twelve and twenty-nine of them share a top count of two occurrences in 400,000 draws, and the walk

follows whichever has the largest numeric value. **A comparison between peak-steered and tail-steered walks is a comparison between two tiebreaks from round one on.**

What remains. The peak *value* is unaffected, since a tie does not change the height of the maximum, only which delta carries it. The compound figure stands. The strategy comparison does not.

The subtraction. Count the ties at the argmax and print the count beside it.

2.14 A RESULT THAT EXISTS AT ONE SEED (OPEN)

Thirty-two of the fixed seeds in this work are the same constant. A statistic appearing at one seed and not at others is a draw, and not a finding.

The subtraction. Sweep the seed and report the spread across seeds beside the headline. The detection floor is then measured instead of derived.

Why it counts here. The three-arm compiler audit establishes that 19 measurements are deterministic: the same seed gives the same answer under three different builds. Determinism is not stability, and the audit does not test whether the answers move when the seed does.

2.15 THE SAME FORMULA WRITTEN TWICE (BUILT)

Two copies of one statistic, differing in a case one of them handles and the other does not.

Instance, in the sieve's own instrument. Three copies of the Rényi deficit sum: the general one over bins, an inline one over the count histogram, and a third inside the Arikan block. The empty-bin rule is opposite at order one and at every other order. A bin holding nothing contributes exactly -1 at a positive order and exactly 0 at order one, since the count in front of the logarithm goes to zero faster than the logarithm goes to minus infinity. The general copy knows both rules. The histogram copy applies the first rule to order one and puts **0.2965 in a table where 0.8272 belongs**, on the headline full-domain run.

A second instance, one layer out. A Poisson prediction at order one missing a $-\log_2(\text{mean})$ term. That term is exactly zero when the mean count is one, the full-domain case. The wrong formula **agrees perfectly at the only point it is read at** and disagrees everywhere else. A defect that hides at the operating point is the worst kind, and only a run at a smaller domain exposes it.

The subtraction. One function that knows the empty-bin rule for every order, with every path going through it. Not a check on the copies: the removal of the copies.

A limit. The reimplementation in another language against another libm carries the same missing $-\log_2(\text{mean})$ term. An independent implementation written by the same author reproduces the author's misconceptions faithfully; what it catches is transcription and folding, and not a wrong idea. What catches the wrong idea is running at a domain where the term does not vanish.

2.16 WHAT THE SIEVE REMOVES

mode	detector	status	what it removes
1 absence as measurement	exact reachability	**built**	3 defects
4 over-reading the output	write mask	**built**	75% of a one-round read
8 optimizer	three-arm build diff	**built**	3 performance claims; 19 cleared
14 one seed	seed sweep	**built**	a survival result confirmed not a draw
15 formula written twice	one deficit function	**built**	2 wrong Rényi numbers
2 hand null	Π_Σ permutation	partial	a factor of 4,200 from the headline
13 tied argmax	tie count at the peak	partial	the walk's strategy comparison
6 estimator ceiling	ceiling beside estimate	partial	the order-1/2 plug-in as a measurement
11 partial for whole	lead with the whole	partial	five reduced-round claims
7 degenerate pairs	zero-difference counter	partial	a false 0-bit output distance
3, 5, 9, 10, 12	none	**open**	nothing

Five modes are closed, five are partial, five have no detector. Of the open modes, mode 3 (the swap test) and mode 5 (printing the detection floor before the result) are both mechanical. Modes 9 and 10 are discipline instead of instrumentation and will not yield to a tool.

Mode 15 deserves the most weight, because it occurs inside the instrument built to catch the others. The sieve does not exempt itself. Both of its instances are exposed by running the same code at a domain it was not designed around, the cheapest detector on this whole list: **run every measurement at a second scale, and compare the shape instead of the number.** A statistic that is a property of the function scales predictably; one that is a property of the apparatus does not.

3

AIMING THE INSTRUMENT

This chapter separates the capability of arbitrary-precision arithmetic from knowing where to point it, and states the rules that six instrument faults produce. Nothing here is about hashing.

The arithmetic computes at a width nothing on a desktop is supposed to reach. That capability is separate from knowing where to point it, and pointing it badly produces findings that are faults in the pointing.

3.1 WHAT THE ARITHMETIC IS

Arbitrary-precision integer arithmetic with no floor under it, a device multiply picking its own arm by operand size, and division done by Newton so that multiplies and shifts are the only primitives anywhere. Measured on one desktop:

what	measured
pi to 10,000,000 digits	116.5 s, checked twice against an independent series
a 1,290,000,000-bit product	1.72 s of device time
2.15 trillion SHA-256 evaluations	summarized in 3 MB of counters
26.6 billion multiply-accumulates	3.2 s, once the work is in the shape the card wants

The last line rewards reading twice. The same permutation null runs at 24 draws a minute in Python and 2000 draws in three seconds as a Gram matrix. The card is not the constraint; the shape of the work is.

3.2 READ THE INTERIOR, NOT THE BOUNDARY

A physical system forces a boundary reading: the interior cannot be reached and is inferred from what escapes. Almost nothing the arithmetic is pointed at has that constraint, and treating it as though it does is a blindfold put on by hand.

Pointed at a compression function, it is tempting to read digests while every schedule word, every state word at every round, and every value built inside a round are readable at every instant. The interior says in one run what the boundary cannot say at all: the two nonlinear functions lag the linear ones by exactly one round, because a value like $(e \wedge f) \oplus (\neg e \wedge g)$ does not depend on e wherever f and g agree. A digest sums every path into one number and can never separate them.

Ask what the object's interior is before deciding to read its surface.

3.3 ASK WHAT IS FORBIDDEN BEFORE ASKING WHAT IS FREQUENT

A frequency needs a null, a bar, an argument and a correction. A support does not: a transition either occurs or it never does, and a transition that cannot occur rules out every trajectory requiring it, permanently, with no statistics involved.

Six hundred excitations find 214 of 256 positions unreachable at one depth. That is a harder statement than any p-value in this work, and it takes one run.

Frequencies are what is left when the rules run out. Read the rules first.

3.4 THE POSITIVE CONTROL DECIDES WHETHER A NULL MEANS ANYTHING

Without one, *we found nothing* and *we could not have found anything* print identically, and there is no way to tell them apart from inside the result.

The clearest case: a compression test finds no structure in a hash. Adding an arm for the digits of pi, a sequence with a *kilobyte* generating program by construction, shows pi compressing to 1.0010 against random's 1.0001. The instrument cannot distinguish a known short program from none. Its verdict is empty, and the claim would be a fact about zlib.

Every instrument gets one: pi for algorithmic structure, a planted relation for a relation search, a withheld range for anything selected, a known-answer vector for any reader, planted structure of each class for a scanner. If the control does not fire, the null is about the tool.

3.5 DRAW THE BAR, NEVER DERIVE IT

Every derived threshold in this work comes in too low, seven of seven. The error runs in one direction only, and that earns a rule instead of more care. Deriving a bar means enumerating the sources of variance, and the ones left out only ever *add* variance. The derived value is always the low estimate. A drawn bar has them all whether anyone thought of them.

the seven, by shape
a shuffle of labels returning identical counts every draw, sd 0.00
a bar at two Poisson floors for a statistic spanning three and a half by construction
chi-square's variance through a logarithm: 0.0582 where shuffling gives 0.1015
$\sqrt{2 \ln N}$, the *expected* maximum, which half of all nulls exceed

Resample instead [21]. Shuffle the labels, redraw the counts at the same N and the same bin count, recompute the identical statistic, read the percentile. Where the statistic is a maximum, a minimum or a range this is not optional, because those have no usable closed form.

3.6 MATCH THE NULL TO THE NUISANCE PARAMETERS

A null that does not match reports the nuisance parameter and calls it structure. A lit set's spectrum depends heavily on how many bits are lit, and the weight wanders between 57 and 140 across the rounds being compared. A null drawn at a fixed weight would therefore report that wander as signal at every step.

Hold fixed whatever the statistic is sensitive to and does not mean.

3.7 NORMALIZE UNTIL THE BASELINE IS FLAT

For points with no preferred direction every mode carries the same expected power, and a degree holds $2l+1$ of them. Per-degree power therefore *rises* as $2l+1$ with nothing in it at all. Drawn: the ratio is flat at 3.867 for a weight of 64 and 5.095 for 128, to within two and a half percent across ten degrees, and the two constants stand in the ratio the finite-population correction predicts, 1.318 measured against 1.333 expected.

Plotted raw, every spectrum rises and the rise is entirely mode count. Divided by $2l+1$ the null is a horizontal line, the baseline is one number instead of a curve, and a departure is a departure with nothing to subtract by eye.

Find the form in which the null is flat, then plot that.

3.8 KNOW WHICH CLASS AN INSTRUMENT CAN SEE

Scanners raced against planted structure of four kinds show the limit. The harmonic expansion beats every rival on anything with a shape, 328 standard errors against 14.8 for the next best, and on a forced parity it reads 2.43 against its own 3.75 floor, *below* its noise. Not worse: blind, because the expansion is a linear map of the lit set and a parity is not a linear property of which bits are lit. The scan that catches it is the crudest in the table.

Best instrument is therefore not a property a single transform has. Run one cheap and complete scan over the spatial class and a second over the algebraic class, and report which fired.

3.9 DISTINGUISH THE FORMAT'S FLOOR FROM THE OBJECT'S

A residual *stopping* at 10^{-16} is float64. A residual *falling* with the width belongs to the object. Only fixed point can tell those apart, and the difference is not academic: an invariance measured in a double reports 10^{-14} and cannot say whether the theorem holds exactly or merely sits where a double runs out. Recomputed in fixed point it reads 2^{-60}, 2^{-92} and 2^{-123} at widths of 64, 96 and 128, tracking the width. The invariance is exact.

Measure the same thing at two widths. If the answer moves, it was the format.

3.10 WIDEN EVERYTHING THAT GROWS, AND NOTHING THAT IS DEFINED

A mod 2^{32} ring *is* the function; widening it computes a different one and fails silently. But the statistics computed *over* that ring grow without bound and must not share its width. An accumulator stored in a thirty-two bit integer, in analysis code computing statistics over that ring, sits nine bits from silently wrapping.

The definition fixes a width. Everything else is arbitrary precision.

3.11 THE INSTRUMENT IS THE FIRST SUSPECT

Six instrument faults, every one producing something that looks like a finding, every one detected by the result being too clean and not by care beforehand:

what it looks like, and what it is
a thirty sigma excess that is a generator with a period of 256
twenty hours of geography that are twenty random trough hours
a swing of 1.394 that is the null's own largest
a band at 280 seconds that is the top of a sorted list
a chirality result fixed by geometry before the data is read
an invariance that is a scalar hiding the structure it sums over

When a result is surprising, check the tool first, and build the check into the tool so the next surprise has to pass it.

3.12 A MEMORYLESS CORPUS, AND WHAT THAT BUYS

A corpus is memoryless when its items carry nothing about each other: drawing one tells you nothing about the next. Establishing it is worth doing early, because it decides which whole families of method are available and which are wasted effort.

It is established here by direct measurement instead of assumption. Distance-to-target autocorrelation across 200,000 consecutive items is inside the null band at every lag from 1 to 1024. Hill-climbing on that distance loses to random draw by 3.81 standard errors on an equal budget. Not tied, *lost*, because it spends its budget probing neighbors that carry nothing.

3.12.1 *Ruled out, at and above a measured floor*

Every ordering strategy at once. If items are independent, search order cannot change the expected cost, and any cleverness about *which* item to try next is spending effort on a quantity that does not exist. One measurement retires the entire family instead of each member separately, the strongest thing a null does anywhere in this work, and the reason it is easy to quote past its evidence.

Both arms read the digest through the leading-zero count, about two bits of entropy per evaluation, discarding everything below the first set bit. That is not wrong, and without a floor it is not bounded either: *we found no gradient* stands in for *no gradient exists* with nothing fixing the difference. Fading an injected gradient until the same statistic loses it puts a number on it: 0.0625 leading-zero bits per single-bit step, with the control flat at every rung. Above that, excluded. Below it, or expressed anywhere else in the digest, it is not looked at.

The engineering conclusion survives intact, because mining's objective *is* that count. The general claim does not. This null carries the most weight here, and it shares the trait of every fault in this chapter: it looks clean, and a readout guaranteeing a near-null looks the same as an object with nothing in it.

3.12.2 *Granted: the arms*

Independence lets observations be pooled by addition with no calibration step. A memoryless corpus can therefore be sampled in any order, at any depth, by any number of observers, and combined exactly.

This half is the useful one and it is easy to miss. Because each arm carries its own exact zero, a count c out of N deviating by exactly $2c-N$, arms of *different* depths pool correctly, the deviations add, the variances add with them, and there is no fitting anywhere. Nothing has to be planned in advance: short arms while exploring, long arms where something looks interesting, all combined into one exact estimate.

Memory per arm is flat. The counters are the same size whatever the depth, and a thousand arms therefore hold 2.15 trillion evaluations in three megabytes, about 700 million per kilobyte, and the footprint does not grow. That makes *keep looking until the question is answered* a real option instead of a wish: the record never fills.

And the two things arms buy do not scale alike, which decides how many to run:

what it buys	how it scales
pooled depth	as $\sqrt{k}$, the slow axis, traversed at whatever rate the device runs
agreement	as 2^k, each further arm halving the chance a run of agreeing signs is coincidence

The second saturates. At 32 arms a unanimous reading happens by chance about once in ten million over 256 cells, and past that point more arms buy depth only. Thirty-two is the knee.

3.13 ELISION, AND HOW TO USE IT INSTEAD OF FIGHTING IT

Elision is a value ceasing to depend on an input, the state becoming identical whether that input was one thing or another. It is normally described as loss, and from the inversion side it is. From the measurement side it is the most useful thing in the machine, because an elided dependency is a *hard* fact about the object and costs nothing to establish.

The support measurement is elision read forwards. Six hundred excitations, and 214 of 256 positions never move at all, each one an elision, each one ruling out every trajectory that would require it, and none of them needing a null or a bar or a p-value. A frequency needs all three and most come back empty anyway.

Three ways to use it:

As a constraint, and not a loss. Where a dependency has been elided, everything downstream of that dependency is decided. That prunes searches over trajectories outright instead of probabilistically, and it makes a support measurement worth more per run than any frequency.

As a clock. Elision accumulates at a rate, and the rate is a property worth knowing. Here the support grows about nine positions a round and closes near round thirty of sixty-four, saying the object spends half its work reaching the point where everything can affect everything and the rest past it. That number is invisible to any reading of the finished output.

As the thing that tells you where to stop measuring. Once a dependency is elided no instrument recovers it, however precise, because the state is genuinely identical either way. A

measurement aimed past the elision point is measuring nothing and will return a floor that looks like a result. Knowing where elision completes is knowing where the readable window closes, and in the case measured here it closes at round thirty while the tempting place to look is round sixty-four.

The general form: **elision is where the object stops being able to tell you something, and finding that boundary is cheaper and harder-edged than anything on either side of it.**

3.14 THE ORDER THESE GO IN

1. State what the object permits before measuring what it does.
2. Read the interior if the interior is readable.
3. Run the positive control. If it does not fire, stop.
4. Draw the bar. Match the null to the nuisance parameters.
5. Normalize until the baseline is flat.
6. Measure at two widths and see whether the floor moves.
7. When something clears, suspect the instrument, and only then the object.

4

THE DIMENSIONAL TRANSFORMS

This chapter states what each transform in this work translates between, what the translation buys, and the single condition under which any of them helps at all. The transforms are the bitpack, fixed point, the number-theoretic transform, the four-step fold, the golden placement, the harmonic expansion, and the permutation read as an orthogonal map.

Seven transforms carry this work. They look unrelated and they do one thing: they express the same information in coordinates where a question is easier to ask. No transform here creates information or destroys it. Each has an inverse or is lossless at full rank.

4.1 THE SEVEN

The bitpack. An integer and a run of 32-bit limbs are the same bytes named twice. Converting one to the other is not a computation; the two are two vocabularies for one object. This lets a number be handed to a device that only knows arrays, without anything being computed.

Fixed point. A real to whatever depth is paid for, stored as an integer scaled by a power of two. The reals are the human coordinate and the integer is the machine's, and the translation is exact in both directions. The whole floor question turns on this: a residual stopping at 10^{-16} is speaking float64, and one falling with the width is speaking the object.

The number-theoretic transform. Coefficients to evaluations and back. A polynomial of n coefficients evaluated at n points costs n^2 pointwise and $n \log n$ through the transform, because the transform shares work between the points that pointwise evaluation repeats.

The four-step fold. One dimension of length N becomes two of $P \times Q$, with a twiddle between them, as in Bailey's four-step transform [1]. Nothing is lost, since it is a relabeling of the same indices, and it lets a transform too large for one pass fit in two.

The golden placement. An index becomes a direction. Index k goes to height $1 - 2(k + 0.5)/n$ at longitude $k\gamma$, the Fibonacci lattice on the sphere [11], and 256 bit positions become 256 points on a sphere with no seam and no pole pile. A one-dimensional list becomes a two-dimensional surface.

The harmonic expansion. A set of points on that surface becomes $(L+1)^2$ coefficients sorted by angular scale. At full rank it is lossless, making it a change of vocabulary and not a summary, and it buys the sorting: structure at a scale concentrates in one band instead of smearing across 256 numbers.

The permutation as an orthogonal map. k observations of a quantity become a vector in k dimensions, and correlation becomes the cosine of an angle. A permutation is then a rotation

within the coordinate subgroup, preserving every length and randomizing every angle. Shuffling the data is this rotation.

4.2 WHAT THEY PRESERVE AND WHAT THEY MOVE

Every one preserves the information and moves the coordinates. That family resemblance has a consequence worth stating flatly: **a transform cannot make a question answerable that was unanswerable. It can only make an answerable question cheap.**

The number-theoretic transform makes the arithmetic faster and leaves its capability where it was. The harmonic expansion makes the boundary reading legible and adds no information to it. Nothing in this work computes anything a slower arrangement could not compute, and stating that plainly keeps the rest of the claims defensible.

4.3 THE SINGLE CONDITION: RANK DEFICIENCY

A transform pays when the object has fewer coefficients than it has points. That is the entire mechanism, in every case.

Multipoint evaluation is fast because a degree-n polynomial has n coefficients however many places it is evaluated. The work at one point overlaps the work at the next. Harmonic sorting is useful because a smooth field has most of its power in low degrees, and the high ones can be dropped. The arms pool because independent observations share a mean, and k of them carry one number's worth of uncertainty instead of k.

That fixes when a transform is useless, and the test for it is direct. A Boolean function of degree below k sums to zero over any k-dimensional affine subspace, the property higher-order differentials [14] and cube attacks [7] exploit, and a cube of inputs therefore collapses if and only if the function is rank-deficient in that sense. Measured over 126 cells at every depth from eight rounds to sixty-four, nothing collapses: every cell sits near 128 of 256, which is full degree.

An object at full rank in a basis has as many coefficients as points, and no change of basis helps. Not *the transform is slow*: there is nothing for it to share. No transform in this work can therefore shortcut a hash, and every one of them speeds up arithmetic, because polynomials are rank-deficient by construction and a hash is built specifically not to be.

4.4 WHERE A TRANSLATION PAYS, AND WHERE IT DOES NOT

It pays. The permutation-as-rotation reading explains why independent observations discriminate so sharply, and explains it better than a counting argument: in k dimensions random vectors are nearly perpendicular, with typical cosine $\sqrt{2/\pi k}$, giving 0.051 at 245 dimensions and 0.025 at 1000. The discrimination is concentration of measure, and not sample size, and that framing predicts the right scaling where counting does not.

It pays. The harmonic expansion beats every rival scanner on spatial structure by a factor of twenty-two, 328 standard errors against 14.8, because sorting by angular scale gives the coordinate in which a shape is one coefficient instead of 256.

It does not. The same expansion reads 2.43 against its own 3.75 floor on a forced parity, below its own noise. A parity is not a linear property of which points are lit. The expansion is blind to it by construction, and the crudest scan in the comparison catches it. A transform translates into one language and is deaf in every other.

It does not. Reading a state through a mirrored placement looks like a chirality test and is a geometric identity: reflection negates every longitude, conjugating every coefficient and negating every phase, for any data whatever. The answer is fixed before the data is consulted. A transform's own symmetries can produce a result saying nothing about the object.

4.5 THE PRACTICAL FORM

1. Ask what coordinate the question is naturally cheap in.
2. Check the object is rank-deficient in that coordinate, because otherwise the move is free of benefit as well as free of cost.
3. Apply the transform, which changes nothing but the vocabulary.
4. Check the result is not a symmetry of the transform instead of a property of the object. Run the same reading on data with nothing in it and see whether the answer moves.
5. Translate back, and state the finding in the coordinate the question was asked in.

Two of the instrument faults in this work fail step four.

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